

Prior Designs & Iterations

2022 RE-RASSOR Original Arm Team

Project Accomplishments:

The 2022 RE-RASSOR team was the first team tasked with creating a robotic arm to manipulate 3D printed pavers for the EZ-RASSOR rover. Like our own team, they were tasked with creating a low cost, almost entirely 3D printed arm to be mounted onto the rover to complete various tasks. This team focused on the software side and developed a complete software architecture around the robotic arm. They implemented a ROS-based control framework, with an in-depth simulation, and investigated computer vision techniques using NVIDIA Jetson Nano and cameras. Similar to ours, they also created an application to control the arm through the Raspberry Pi & Arduino. While this project has a larger scope than ours and was ambitious and successful in what they were trying to achieve in the software side, their physical design had significant mechanical challenges that caused the arm to be unable to integrate into the RE-RASSOR platform.

One of the primary limitations that the 2022 team had was the significant emphasis on the software portion of the project, rather than the mechanical design and robustness needed to manufacture a mechanical arm. While their simulation showed good results, the translation to a real design proved to be much more difficult.

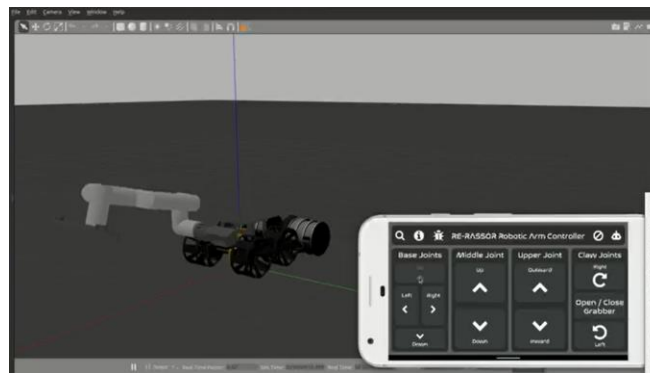


Figure 1: 2022 Preliminary Simulation

Limitations:

The biggest issue this team faced, and one that heavily influences our current design, is the weight of their proposed arm. The arm relied on several stepper motors distributed across the joints of the

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arm to control it. As mentioned in the section about the 2025 RE-RASSOR team, stepper motors are very powerful but much heavier than the BLDC motor design. As the arm extended outward, each joint was forced to support not only its payload but also the weight of every motor and structural component farther along the chain. Since the motors were so heavy, there were significantly higher torque requirements at the base joints of the arm, which reduced the actual lifting capability. This is something the team ends up identifying in their final design document.

The team also experienced difficulties transitioning from software simulations to hardware implementations. The team mentions that while they were successful in simulating the arm's motion, integrating the hardware, electronics, and controllers introduced further challenges that didn't have enough time for a thorough implementation. They recognize that if they had validated their hardware design early on before focusing on the simulations and complex software, this may have prevented many of these integration problems for them.

Lessons Applied to our Design:

The work done by the 2022 RE-RASSOR team provided us with valuable lessons that directly influenced some of our design decisions for our arm. One of those decisions was to keep our arm as light as possible using the BLDC motors from the 2025 team. Since the weight of the arm required not only the base but the joints as well to produce high torque, we wanted to avoid this scenario if possible. Learning from the 2022 design, we plan to use a stepper motor for the base of the arm connecting to the chassis and using BLDC motors for the joints of the arm. This way, we can still use the power of the stepper motor at the base to rotate the whole weight of the arm, and it keeps the weight on the low side.

Another design decision that we are going with is to keep the 3D print infill percentage on the low side. When looking at the arm design from 2022, one thing that stood out to us was the infill on the arm pieces, as they were very high percentages and some cases 100%. This indicated to us that we want to avoid using such a high infill for arm pieces and keep heavier pieces for the base of the arm. In a similar way to using the BLDC motors instead of stepper motors for the joints, this allows us to keep the base strong but the arm itself light.

Finally, their team showed us the importance of focusing on the hardware of the project, instead of diving headfirst into the software of our mechanical arm. While their software was extremely impressive, especially their simulations for their whole entire bot, it led to a lot of time being spent on the software, when some of it could've been spent on validating the hardware design. Learning from their

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experience, our time decided to dedicate two members of our team to working on our hardware design, with our PM assisting us until we are confident in the arm design and it has been tested. This way, we can have software and hardware design happening at the same time and produce prototypes as soon as possible for software to start messing with.

In summary, while the original RE-RASSOR arm did not fully achieve its intended goal, it established an invaluable technical foundation for our team. By analyzing both its successes and its limitations, our team has been able to develop a more practical mechanical design centered on lightweight construction, brushless DC motors, and ensuring that we spend a significant amount of time vetting our design.

Credits:

Credits for the work of the 2022 RE-RASSOR group go to:

Project Sponsor(s):

- Mike Conroy
- Florida Space Institute (FSI)

Individual Team Members:

- Matheus Pavini Franco Ferreira
- Jack Bailey
- Braden Steller
- Israel Booth
- Kartik Rana

2023 RE-RASSOR Extension Team

Project Accomplishments:

The 2023 RE-RASSOR Extension team focused on expanding the software of the existing RE-RASSOR arm, rather than redesigning the mechanical structure. From their GitHub repository, they

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primarily focused on integrating modern ROS2 into the RE-RASSOR software infrastructure, and designing the remote-control interface, a mobile app, used to control the rover.

The largest accomplishment in software development they did for the RE-RASSOR is to refactor using ROS2 Humble. This modernized and improved the earlier software architecture for the rover itself. In addition, they also created updated ROS packages, scripts to launch the rover software and systemctl services for the rover. These improvements created a more maintainable software platform for future development.

The next accomplishment was the creation of a React Native application to control the RE-RASSOR rover. Previously, the team at FSI had to use a mobile app, which had issues due to apple requiring constant updates to stay on the app store. This currently controls the rover and has the opportunity to swap into different control modes, which is perfect for our project. Their modular design allows us to easily add new commands to the web application without significant architectural changes to the whole application.

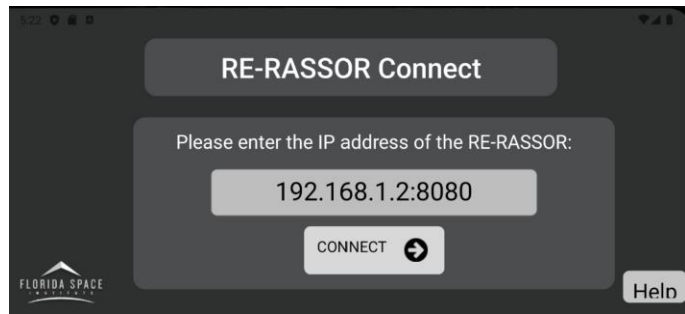


Figure 2: 2023 Team's Rover Connection UI

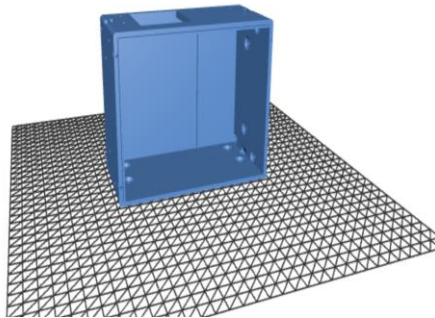


Figure 3: Current RE-RASSOR Body in CAD

Mechanically, the team worked to bring the RE-RASSOR rover into its current state. This includes all the current CAD models on the RE-RASSOR rover, as well as updating the old EZ-RASSOR rover with a new shoulder design. Because this team mostly worked to improve the software on the rover, most of the borrowed and referenced work will pertain to their software.

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Effects for our Project:

This team's work directly influences most of the software development we have to do for the arm design. Since this team set up the entire HTTP request system through the localhost application, our team is deciding to build that by adding an arm module for our robotic arm. This will be an additional page reusing and repurposing existing code for the base of the rover. Similarly, the ROS2 communication framework can be extended to support our new BLDC motor controllers we plan to use from the 2025 RE-RASSOR team. While the underlying hardware differs between the two, using the 2023 team's work with ROS2 and the rest of the software on the rover, we will just need to add new modules dedicated for arm control.

In relation to our stretch goals, the extension team also implemented the groundwork for a camera subsystem on the RE-RASSOR rover. What they wrote enables camera streaming from the rover that can detect ArUco markers in the lab. This is great for our design, as we can use their camera module to add functionality to move the arm to a certain offset from these markers and/or other movements. From their work, we don't need to focus on writing a camera system from the ground up and instead can immediately start leveraging a native library for RE-RASSOR.

Credits:

Credits for the work of the 2023 RE-RASSOR group go to:

Project Sponsor(s):

- Mike Conroy
- Florida Space Institute (FSI)

Individual Team Members:

- Noah Gregory
- Riya Singh
- Adam Whitlock

2025 RE-RASSOR BLDC Motor Team

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Project Accomplishments:

The 2025 RE-RASSOR BLDC motor team successfully redesigned the drivetrain system of the Florida Space Institute's RE-RASSOR rover by replacing the previous stepper motor design with a well-suited brushless DC motor system. The primary objective of this was to decrease the rover's weight and improve energy efficiency while maintaining the original rover's architecture. This caused the new design to focus on constant rotational control rather than power as with a stepper motor. Since our project relies on us using their BLDC motor design for our arm, this assists us greatly with the nature of building a mechanical arm requiring precision over power.

Beyond a prototype, the 2025 RE-RASSOR team produced extension documentation including CAD files for all components, large documentation regarding CAD files for all components, large documentation regarding process, materials, and project history, allowing us to adapt their work to our own seamlessly.

Mechanical Design & Process:

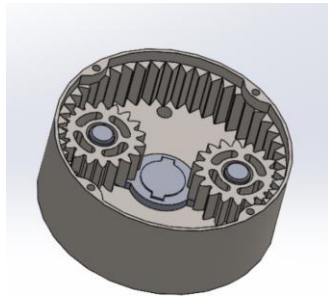


Figure 4: RE-RASSOR CAD Gearbox

While the primary focus for the 2025 RE-RASSOR team was creating a new motor for the rover to use, and controlling it, the team also accomplished significant mechanical integration with the current RE-RASSOR design. Instead of redesigning the whole entire rover to fit their needs, they took a modular approach and designed the BLDC motors to be compatible with the current chassis and other 3D printed parts. This is due to the project being intended for additive manufacturing, allowing students or researchers to make additions to the rover without friction.

Their design process involved evaluating motor mounting methods, configuration of their gearbox, and alignment of the shaft of the motor with the rest of the mechanical components. The team compared a direct-drive and geared-drive solution for their gearbox before recommending planetary gearboxes or multi-stage spur gear reductions where additional torque was required. This means that their design can be adapted to scenarios where more torque is required, or resources are scarce, and the cost must be cheap, allowing students and teams like ours to adapt their design to our needs.

Another important contribution was maintaining compatibility with the rover's existing 3D printed chassis. Rather than redesigning the entire vehicle, the team modified only the parts necessary to

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accommodate the lighter BLDC motors. This reduced manufacturing costs and ensured that previously printed rover components could still be reused. In addition, the team worked to ensure that materials needed for constructing the gearboxes and motors themselves used readily available materials at the FSI lab and already being used in other places of the rover. This makes it much easier and cheaper to manufacture the rover, and any additional addons we need.

Software Design & Development:

The software developed focused on supporting the mechanical design goal of implementing the BLDC motor design, which is allowing for efficient rotational control of the motors, while maintaining compatibility with the rest of the RE-RASSOR rover's architecture. To accomplish this, the team continued the use of the Arduino environment's c++ library to get low level access to the hardware.

The software is designed to interface with the main controller for the rover, which then passes commands to the Arduino Mega, which processes and generates PWM signals to control the motor. A major design goal was maintaining constant motor speed despite changing mechanical loads. To accomplish this, the software supports sensor feedback from Hall sensors or rotary encoders, allowing the controller to continuously monitor motor position and speed. This feedback enables more stable motor performance under varying operating conditions.

Due to the communication protocol mirroring the original stepper motor interface, the high-level rotor software did not require major refactoring to accommodate the new drive system, which reduces the effort to integrate the BLDC motors, and will allow for future projects utilizing their work to require few code changes.

Effects for our Project:

Due to the nature of our group's work, the effort made by the 2025 team will greatly influence our hardware and software development and design. Since the goal of our project is to design a robotic arm that *must* use this team's BLDC motor design, their work directly ties into what we need to do. From the creation of their design, it drives us to utilize their motors in our design due to their lightweight design, and strong control of the motor's rotation. Since a mechanical arm must be lightweight and stable, the BLDC motor created by the 2025 RE-RASSOR team gives us an immediate solution to two core challenges we would have to face. Without their design, we would either use the previous stepper motor design and integrate it into our arm, leading to heavy weight and too much power, or create an entirely

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new motor and gearbox design, significantly increasing the scope of our project. Therefore, we can focus on the actual architecture of the arm and use their gearboxes and BLDC motors when needed. As shown later in the document, there are a few key places that they will be placed to control the joints on the arm, and since they are already integrated with the rest of the rover, it is easy to build an arm that utilizes both the BLDC motors and a stepper motor for the base.

Their discussion and research into gearbox selection is valuable to us because robotic arms require high joint torque with precise control. Although our application emphasizes positional accuracy rather than continuous vehicle motion, the tradeoffs between direct drive, planetary gearboxes, and spur gear reductions remain nearly identical to the 2025 team's study. Their analysis provides useful guidance when determining the appropriate transmission for each arm joint.

In terms of their software driving our design, our team will be interfacing with their code to control the BLDC motors over HTTP requests like the drive controls on the rover. Since their software is modular and easy to use, will have their packages as a dependency and modify the rover Arduino code to communicate with our arm subsystem. This allows us to focus on a high-level software design, without needing to delve into the low-level details involved in programming the motors from scratch.

Overall, they establish a strong baseline for our design for which our team can start our robotic arm, without worrying about hardware implementation. This way, we can focus entirely on the mechanical design and communication between the localhost server and the rover.

Credits:

Credits for the work of the 2025 RE-RASSOR group go to:

Project Sponsor(s):

- Mike Conroy
- Florida Space Institute (FSI)

Individual Team Members:

- Alex Fowler
- Nicholas Bernhoft
- Gabriel Pullam
- Noah Fourari

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- Jake Weber
- Kenneth Arias

Inspiration and Adaptations to our design

When researching mechanical arm designs, we took into consideration many factors that we needed to satisfy, such as degrees of freedom, weight, and overall control of the arm. One completely open-source design we found gives us a large foundation that we can work on to adapt to the RE-RASSOR rover. That design comes from the [EEZYbotARM](#) design. This project provides a fully open-source, desktop-scale robotic arm designed to be inexpensive, easy to manufacture, and almost entirely fabricated with common 3D printing materials. Like our initial sketches and ideas, this arm is a 3 degree of freedom arm (excluding the base), that utilizes modular and cheap components for their final product. Since all the CAD models are publicly available, it is a popular design for robotics projects to adapt an arm design to suit their needs.



Figure 5: EEZYbotARM Sample Arm

One of the best parts of the EEZYbotARM is its emphasis on additive manufacturing. Nearly every structural component is designed to be 3D printed, including the base, shoulder, elbow, wrist supports, motor mounts, and linkages. This dramatically reduces manufacturing costs while allowing damaged parts to be quickly reprinted. The CAD models are organized into modular assemblies, making it straightforward to modify individual joints without redesigning the entire arm. The printed components are lightweight in their design as well. The arm also incorporates printed bearing supports and mounting surfaces that simplify assembly using commonly available hardware that is available in most labs.

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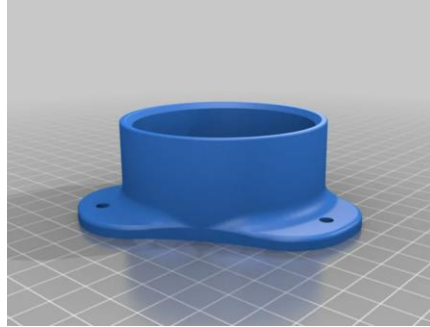


Figure 6: EEZYbotARM Base

Effects on Our Design:

The EEZYbotARM serves as the mechanical foundation for our robotic arm but will require several significant modifications to satisfy the requirements of the RE-RASSOR project. The first core modification is completely overhauling the motor drive system to use the required BLDC Motors. This will help us in the end, as we would end up needing more strength compared to the small servos powering the arm design here due to the scaling. This will also require us to rewire and ensure that the EEZYbotARM design can interface well with the electrical components required to move each of the BLDC motors in our arm. In addition to using these motors to drive the arm, we are deciding to go with a stepper motor to power the base of the arm. This is because the EEZYbotARM modifications we will make will cause the arm to be much larger in scale, so we will need a strong base to generate enough torque for the arm.

Another major modification we will need to make is the scaling and reworking of many of the components in the arm design. Because this arm was developed as a standalone system, it is not meant to interface with other pieces of hardware and/or a chassis like the RE-RASSOR rover. Not only will we need to integrate a modified version of the arm into the front of our rover, but we will also need to scale the arm's design. Our application requires a larger workspace and increased reach, so many of the structural components will need to be resized.

A final effect on our design is the reinforcement of a claw/gripper design for the end effector. In our original sketches and talks, we wanted to go with a claw that could pick up the 3D pavers as specified. In seeing this design, it reinforced the idea that it is feasible and acted as proof of concept for other items.

In the EEZYbotARM, the gripper is designed horizontally to pick up small items such as a ball. A new end effector will need to be developed to better accommodate the objects and tasks expected during operation (such as our pavers) while maintaining compatibility with the wrist assembly.

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Overall, while many significant design changes are needed, the EEZYbotARM project acts as a great starting point for us as all the CAD files are publicly available, and we can see proof of concept of what we need to build for the rover.

Credits:

The EEZYbotARM is an open-source robotic arm developed by Brian E. Siepert ([daGHIZmo](#)).

The original resources include

- Original CAD models for all components that make up the EEZYbotARM
- Assembly documentation
- GitHub Repository containing documentation

Our General Design

Deliverables & Objectives:

The main objective of our team is to design, build, test and deliver a fully functioning, scaled robotic arm that integrates with the existing RE-RASSOR rover platform. As this project is built on multiple years of earlier student development, our baseline objectives are heavily weighted toward developing a solid, physically reproducible mechanical design while building out the web-based software framework to drive the arm.

Minimum Viable Product/Baseline Objectives:

To fulfill our contract, the completed system must achieve the following milestones:

- **Physical Manipulation Capabilities:** The robotic arm must be mechanically capable of autonomously or manually targeting, lifting, moving, and precisely positioning the project's standard 3D printed pavers. These pavers weigh under one pound with diverse shapes and sizes. The arm must execute these tasks without causing structural deformation to itself or the rover chassis.
- **Rover Stability Maintenance:** An important performance objective is that the arm must operate within its specified envelope without causing the RE-RASSOR rover to flip, tip, or lose traction. Because the base body of the rover (excluding wheels, lid, and electronics) weighs not a whole much due to it being made out of lightweight PETG plastic, any

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extension of the arm shifts the center of gravity drastically. The arm's weight and motion profile must be balanced to maintain steady operations.

- **Control System Integration:** The rover control system must be easy to set up, visualize, understand and use. And must incorporate arm control systems together seamlessly with the rover's base control system without causing system to back up or reduce performance.

Detailed Technical Deliverables:

When the project is finished, the team will give the sponsor a complete handover package. This package, which is divided into four different categories, guarantees complete continuity for upcoming student teams:

Computer Aided Design (CAD) Packages:

- **Fully Realized 3D Assembly Models:** Complete. STEP and .STL files detailing every link, joint, bracket, and gear assembly of the robotic arm.
- **Rover Modification Models:** CAD files detailing the exact geometry of any custom supplemental mounting plates or counterweight brackets affixed to the rover body.

Software and Firmware Source Code:

- **Web Interface Upgrades:** Clean frontend code integrated into the existing web application UI that features intuitive controls for managing the arm's multi axis movements seamlessly alongside the primary drive controls.
- **HTTP Endpoint Registry:** Well documented backend application routes managing incoming HTTP requests assigned to arm actuation.
- **Low Level Motor Control Logic:** Optimized C/C++ Arduino sketches managing the distribution of pulse width modulation (PWM) or serial commands to the custom Brushless DC Motors (BLDCM) and tracking rotary feedback. All source code will be securely tracked across the designated GitHub branches.

Fabricated Physical Hardware:

- **Functional Arm Assembly:** A fully manufactured 3D printed mechanical arm primarily composed of PETG plastic to provide structural endurance against environmental heat loads.

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- **Electromechanical System:** An integrated suite consisting of lightweight Brushless DC Motors, associated gear reduction units (such as cycloidal or worm gearing), complete power wiring harnesses, and local network microcontrollers.

Engineering Documentations:

- **Assembly Instructions:** Step by step documentation detailing physical assembly, fastener sizing, wire routing constraints, and maintenance requirements.
- **Verification and Validation Reports:** Documented logs of unit tests, full system integration checks, and formal acceptance testing witnessed by the project sponsor.
- **Future Work Backlog:** A clear record of failed design iterations, lessons learned, and recommended improvement tracks for subsequent teams.

Scope of Engineering Investigations & Information Targets

To start actually turning initial conceptualization to physical assembly, the team has outlined several critical engineering metrics that must be calculated or verified before finalizing the design:

Required Metric / Information Target	Operational Importance and Design Impact
Maximum Actuator Load Capacity	Finding the precise torque limits of a single BLDCM to determine if multi motor clustering or heavy gear reduction is mandatory at high moment joints like the shoulder.
Wiring Envelope Tolerances	Determining the exact minimum and maximum lengths of flexible wiring required to route power and signals safely through moving joints without pinching or binding.
Chassis Structural Thresholds	Measure the wall thickness and material strength of the inherited rover body panels to have save mounting load limits.

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Empirical Base Weight Matrix

Record precise weights of the bare rover chassis, electronics, battery, and track system to accurately calculate counterweight ratios.

Scalability Tracks and Stretch Goals

If the baseline MVP specifications are locked in and validated early, the scope of the project will expand to include analysis of specific, highly desirable stretch goals approved by the sponsor:

Modular / Interchangeable End Effector System:

Redesign the last wrist joint to use a mechanical quick detachable joint. The baseline hand is designed specifically for 3D printed pavers, but the rover can be outfitted with a modular tool changer to quickly change to other tools (for example, specialized scoops or regolith collections bins).

Additional Wrist Rotation Axis (Roll Control):

Introduce a secondary actuator within the wrist assembly to expand the mechanism to 3 degrees of freedom, granting full rotational control along the roll axis. This eliminates a huge limitation of the baseline pitch only wrist, allowing the rover to grasp asymmetric or side targets. Risk Note: This adds significant weight and complex wire routing to the end of the arm.

Computer Vision & Autonomous Object Pickup:

Eliminate manual driver intervention by integrating an extra ArduCam unit onto the forward assembly or arm hand. The software subsystem will ingest video frames, run local object detection or ArUco tag identification routines, translate pixel coordinates into real world workspace coordinates, and command the drive and arm systems to perform fully autonomous paver retrieval.

Mechanical Design

Base Concept:

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This preliminary concept illustrates the team's initial approach to integrating a robotic arm with the existing RE-RASSOR rover platform. Rather than designing an entirely new rover, the concept focuses on modifying the current platform while preserving as much of the existing drivetrain and software architecture as possible. The arm is intended to provide sufficient reach to manipulate the project's pavers while minimizing interference with the rover's existing systems approach to integrating a robotic arm with the existing.

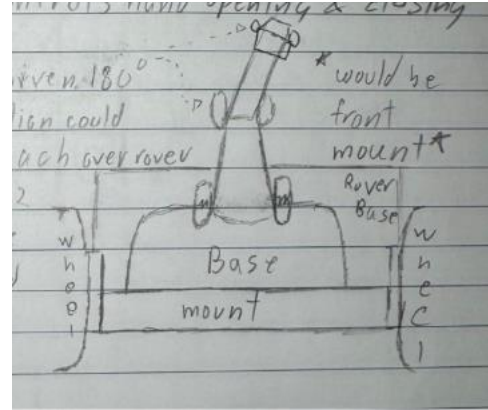


Figure 7: Front View Sketch of Arm Design

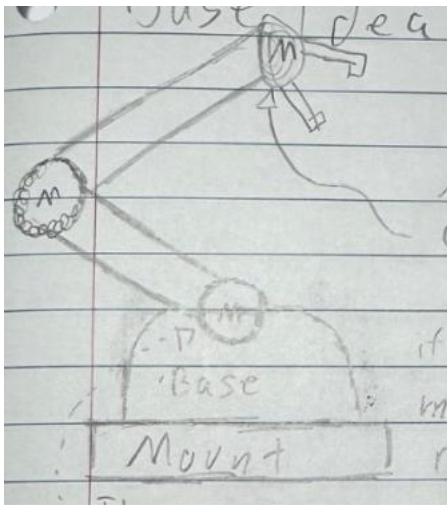


Figure 8: Side View Sketch of Arm Design

The initial design utilizes a multi-jointed arm driven primarily by Brushless DC Motors (BLDCMs). Each major joint is responsible for a single degree of freedom, allowing the arm to extend, raise, lower, and position the end effector. The concept also identifies the possible need for multiple BLDC motors at the shoulder joint depending on the final payload capacity and torque requirements determined during future testing. Since the shoulder joint experiences the greatest bending moment while lifting objects away from the rover, additional actuation or gearing may be required to safely support the arm throughout its operating range.

Mounting System:

One of the primary design considerations identified during early brainstorming was the mounting location of the robotic arm. The concept proposes a dedicated mounting platform attached directly to the existing rover chassis rather than replacing major structural components. This approach minimizes modifications to the inherited rover while allowing future teams to remove or redesign the arm without affecting the remainder of the platform.

The mounting system will likely house the primary rotational gearbox as well as the motor responsible for rotating the arm about its vertical axis. Locating the gearbox within the mounting assembly reduces the amount of moving mass located farther from the rover and allows structural loads to be transferred directly into the chassis. The team also identified that the current rover mounting platform may require extension in length to properly support the arm and distribute loading over a larger area.

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Determining the minimum amount of modification necessary while maintaining sufficient rigidity will be one of the primary engineering challenges during prototype development.

The current mounting platform design consists of an $8 \times 8 \times 1$ inch base plate that serves as the primary interface between the rover chassis and the robotic arm assembly. The platform incorporates dual mounting flanges containing three M5 clearance holes per side, allowing it to attach directly to the existing RE-RASSOR mounting points while maintaining compatibility with the

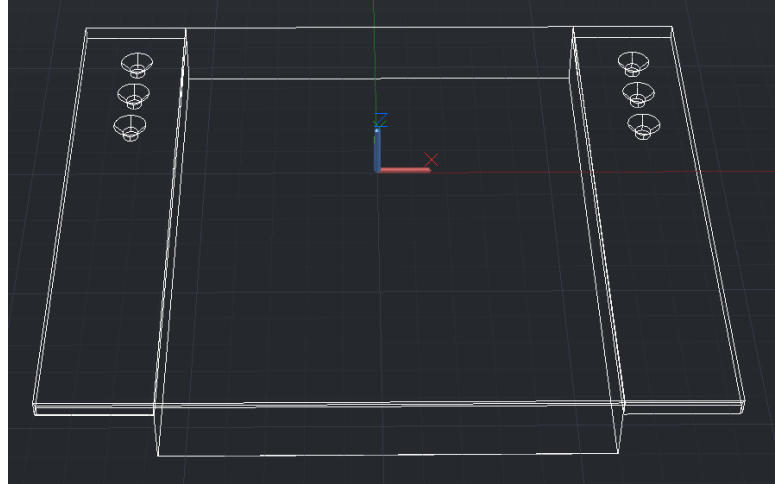


Figure 9: Mounting Platform CAD Design

current rover architecture. To match the existing hardware configuration, each mounting hole includes a countersunk entrance to properly seat the fasteners and simplify assembly. The platform will be manufactured using fused deposition modeling (FDM) with a 35% gyroid infill, selected over traditional rectilinear infill because of its superior multidirectional strength and improved resistance to bending and torsional loads. Although this choice increases print time and material usage slightly, additional structural integrity provides a more reliable foundation for early mechanical testing.

The decision to prioritize structural strength was heavily influenced by observations from previous RE-RASSOR teams, whose documentation identified arm weight and mounting stability as recurring engineering challenges. Rather than optimizing the initial prototype for minimum weight, the current design emphasizes rigidity and durability to establish a dependable baseline for future iterations. Standardized M5 hardware, mirrored mounting geometry, and a symmetric layout were selected to simplify fabrication, assembly, and future modifications while ensuring compatibility with existing rover components. As additional testing is completed, future revisions may incorporate weight reduction features or topology optimization; however, the initial prototype intentionally favors reliability and repeatability to support the project's research-focused development process.

Arm Reach and Workspace:

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The initial concept emphasizes maximizing the arm's reachable workspace while remaining within the stability limitations of the existing rover. One possible configuration identified during brainstorming allows approximately 180 degrees of rotation, enabling the arm to reach both the front and sides of the rover without repositioning the drivetrain. If achievable within weight and wiring constraints, this increased operating envelope would improve the rover's flexibility during future educational demonstrations and simplify object manipulation tasks.

Increasing the arm's reach, however, also increases the bending moments applied to the shoulder joint and shifts the center of gravity farther away from the rover's base. These competing factors will require additional analysis during the CAD and prototype phases to determine an acceptable compromise between workspace and structural stability.

Counterweight System:

One of the most significant concerns identified during early design discussions is rover stability. Previous RE-RASSOR teams experienced balance issues when mounting robotic arms, making weight distribution a primary design consideration. To address this concern, the concept proposes locating a storage bin or counterweight system at the rear of the rover.

The storage compartment could serve multiple purposes simultaneously. Beyond acting as a passive counterweight, the bin may also provide storage for collected pavers or future regolith samples. The concept further proposes the possibility of incorporating a hinged dumping mechanism, allowing collected materials to be emptied without requiring manual removal. By designing one component to perform multiple functions, the team hopes to reduce unnecessary weight while improving the rover's overall capabilities.

End Effector:

The concept includes a simple gripping mechanism designed primarily for manipulating the project's 3D printed pavers. At this stage, the primary objective is reliability rather than complexity. The end effector is intended to provide sufficient gripping force while minimizing overall weight located at the furthest point from the rover.

Future iterations may expand upon this design through interchangeable tool attachments, allowing the same arm to perform additional educational tasks beyond paver manipulation. This modular approach would allow future RE-RASSOR teams to develop specialized end effectors without redesigning the remainder of the robotic arm.

Misc. Mechanical Alternatives:

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There have been multiple alternatives considered for many of the design features described in this document. These alternatives were rejected for various reasons in favor of the chosen design features, which will be discussed here.

There were three alternative ideas considered for the location where the arm would be mounted on the rover:

- Top mounting
- Back mounting
- Double mounting (2 arms mounted in front and back)
- Bottom mounting

Top mounting of the arm was rejected for the reason that it would likely interfere with usage of the power switch and charging port. Both of these components are located at the top of the rover, and would likely need to be moved if the arm was placed there.

Back mounting the arm was rejected since it would require a longer arm to reach the front of the rover. It would also slightly interfere with the fan, as that component is located on the back of the rover.

Double mounting arms on both the front and back of the rover was rejected because it would require us to use twice the number of materials. It would also still interfere with the fan component at the back of the rover. The second arm would also require additional controls on the User Interface, draw additional power from the battery pack, take additional network and processor bandwidth to send commands, and increase the rover's weight. The only benefit of this alternative would be the removal of any requirement for a counterweight, as the arms would balance each other's weights. If this route is taken in our design later it will be due to issues with said counterweight; however, until a design is attempted this approach is deemed as expensive and unsuitable for what is intended to be an affordable and attainable entry into robotic.

Bottom mounting the arm was trivially rejected for the reason that it would reduce the rover's already low clearance and significantly impair movement. It would also require significantly more effort and engineering to create a design that could successfully pick up the pavers.

Another decision which was taken into consideration was the choice of plastic to use in the 3D printed components of the rover. For this, the only alternative that was worth considering was replacing the PETG plastic with Polyvinyl Chloride (PVC) plastic. PVC would be lighter, cheaper, and more rigid than PETG. However, PVC plastic was rejected, as it is significantly more vulnerable to heat. The rover is

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expected to be exposed to drastic temperature changes in its usage, so PVC plastic would quickly fall apart. PLA was rejected for similar reasons as it also has a low melting point and experiences severe warping and melting under relatively low heat. However, its affordability makes it suitable for test prints.

Another important alternative that was considered was the use of stepper motors for the rover's movement and joints. A previous team actually attempted to build an arm using stepper motors and was able to create an almost functional design. However, stepper motors were eventually rejected in favor of Brushless DC motors, as the stepper motors were too heavy for the rover to balance, and caused the rover to tip over whenever the arm was used.

Lastly, multiple alternatives to the cycloidal gearbox were considered for use in the shoulder joint of the arm. These being:

- Worm gear
- Bevel gears
- Screw gears

These alternatives were considered as ways to fix the issue of top-mounting the arm interfering with the power switch and charging port, as they would allow the motor to be placed away from those components. However, the worm gear and screw gears were deemed to be unnecessarily complicated, and the bevel gears would require compromising the torque ratio of the joint, so these options were discarded along with the top-mounting idea.

Software Design

Current design plan for HTTP:

Architectural Overview and Communication Topology

The software control framework of the RE-RASSOR platform employs a decoupled, distributed node communication topology to provide safe isolation of high level remote user interface inputs from low level electromechanical execution. This network architecture employs an explicit HTTP REST API and asynchronous execution threads to avoid blocking conditions on the main telemetry loops.

It begins at the user interface (the React native or web application hosted on a client machine on the local net). The application communicates with the rover or its

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attachments through structured HTTP POST requests over a local network (via a private router) to the rover's main processing board (Raspberry Pi or Le Potato) running on a specific IP address on port 5000.

The High Level Network Bridge: Flask and SocketIO Core

Server.py is the entry point of the microprocessor. It instantiates a multi-threaded web application using the Flask framework augmented with Flask SocketIO. This node is the main gateway from the web to ROS2.

The REST API Entry Point: The root URL path (POST /) functions as a unified command dispatcher. It processes multi faceted incoming payloads containing actions for wheels, shoulders, arms, drums, or autonomous targets.

Payload Verification and Normalization: To enforce system integrity and safeguard physical components from invalid operational states, the server feeds incoming JSON requests immediately into a rigid schema validator. The verification system strictly parses fields using predefined structural constants such as `WHEEL_ACTION_KEY = 'wheel_action'`, `SHOULDER_ACTION_KEY = 'shoulder_action'`, `FRONT_ARM_ACTION_KEY = 'front_arm_action'`, and `BACK_ARM_ACTION_KEY = 'back_arm_action'`. If an explicit action violates defined types, such as passing arbitrary float ranges instead of the strict discrete bounds enforced by ArmAction (`LOWER = -1.0`, `STOP = 0.0`, `RAISE = 1.0`), the server aborts execution, throwing a `VerificationError` and responding with an HTTP 400 Bad Request status code. Validated inputs are mapped directly into structured runtime objects via `command.py`.

ROS2 Workspace Integration and Publisher Instantiation

Once a payload is verified, `server.py` converts the web variables into native ROS2 messages. The server spins up a dedicated ROS2 node lifecycle thread that owns explicit publishers for core motion topics:

- **wheel_publisher:** publishing `geometry_msgs/msg/Twist` to `/ezrassor/wheel_instructions`
- **shoulder_publisher:** publishing to `/ezrassor/shoulder_instructions`
- **front_arm_publisher / back_arm_publisher:** publishing to `/ezrassor/front_arm_instructions` and `/ezrassor/back_arm_instructions`

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By executing the flask thread dynamically as a background daemon. The server guarantees that incoming consumer HTTP requests are captured asynchronously without introducing jitter or latency to the low level ROS2 compilation executor.

The Low Level Serial Translation Layer

- The server.py reads network traffic and converts it to ROS2 topics, but the conversion to hardware voltages is done by a separate downstream bridge script: serial_driver.py. This node is a high frequency subscriber, that subscribes specifically to the server's mechanical Twist topics.
- **Physical Bus Addressing:** The driver establishes serial links over USB buses mapped using persistent symbolic links created in the Linux subsystem (/dev/arduino_wheel and dev/arduino_drum) operating at a standardized transfer rate of 115200 baud.
- **Binary Serialization and Wire Protocol:** Microcontrollers lack the computational efficiency to fluidly ingest full high level string payloads or floating point structures under real time constraints. Therefore, serial_driver.py acts as a data compression encoder, flattening continuous Twist parameters down to single discrete, 8-bit hexadecimal opcode bytes before sending them over the wire. The established binary communication wire protocol maps as follows:

Target Device Bus	Monitored Input Parameter	Computed Hex Opcode	Intended Outcome
/dev/arduino_wheel	linear.x > 0.0	0x01	Forward Drive
/dev/arduino_wheel	linear.x < 0.0	0x02	Reverse Drive
/dev/arduino_wheel	angular.z > 0.0	0x03	Left Spin
/dev/arduino_wheel	angular.z < 0.0	0x04	Right Spin
/dev/arduino_wheel	linear.x == 0 & angular.z == 0	0x00	Full Braking

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/dev/arduino_drum	msg.linear.y > 0.0 (Shoulder)	0x05	Raise Front
/dev/arduino_drum	msg.linear.y < 0.0 (Shoulder)	0x06	Lower Front
/dev/arduino_drum	msg.angular.y > 0.0 (Shoulder)	0x07	Raise Rear
/dev/arduino_drum	msg.angular.y < 0.0 (Shoulder)	0x08	Lower Rear

Transition Strategy: Refactoring the HTTP Schema for the Scaled Robotic Arm

While the inherited HTTP bridge structure functions reliably for the legacy system, it is custom tailored exclusively to support dual drum excavation systems (front_arm_action, back_arm_action, front_drum_action, back_drum_action). Because our current senior design group is pivoting the rover's utility to manipulate solid 3D printed pavers using a single, multi-axis scaled articulating arm, our team must execute an aggressive engineering refactor of this API ecosystem.

Expansion from Binary Opcodes to Direct Kinematic Coordinates

Our new arm relies on lightweight Brushless DC Motors matched with precise magnetic field encoders. A simple binary 'RAISE' or 'LOWER' opcode (0x05 / 0x06) cannot safely coordinate multi joint interpolation or precise endpoint placement. The REST endpoint must be extended to accept absolute geometric position states or joint angles.

Refactoring the Middleware Chain

- **1. request.py Validation Modifications:** We will strip out the binary constraints checking for front/back drums and append an array validator checking for the existence of explicit floating point values for each targeted arm joint angle.
- **2. command.py Translation Enhancements:** The ArmAction class will expand from a basic discrete enum into a comprehensive kinematic state class capable of holding floating point structures.

Integration of Previous Requests

1 Analysis of Inherited Repository Branches

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One of the group's main architectural challenges is to reconcile overlapping and competing codebase frameworks developed by previous teams. Looking at the inherited workspace, the code is organized into two distinct operational pipelines: a localized Python web to serial bridge and a native C++ ROS2 framework leveraging low level serialization or outbound HTTP workers.

The Python Web to Serial Ecosystem: This pipeline utilizes `server.py` to instantiate an inward facing Flask REST and SocketIO web server running on port 5000. It handles remote string based request configurations, parses them through an explicit validation engine (`request.py`), and converts them into standardized ROS2 topics. Downstream, a secondary Python node (`serial_driver.py`) subscribes to these topics and uses the `pyserial` library to write discrete 8-bit binary hex opcodes to two physical USB mount paths: `/dev/arduino_wheel` and `/dev/arduino_drum`.

The Native C++ Client/Driver Workspace: In stark contrast, the second branch has very structured C++ nodes. The `motor_controller.cpp` node is not a server. It is an HTTP client. It creates a local watchdog, and fires off an asynchronous `libcurl` worker thread in the background (`httpWorker`) to push outbound JSON packets away from the ROS node, to a hard coded remote target IP.

2 Conflict Resolution and the Unified Integration Strategy

To scale our robotic arm without breaking system stability, our team needs to have a definitive integration baseline. Keeping the Python Flask server loop and the C++ outbound curl injector with serious network loopback hazards, variable racing, and unnecessary command latency. We have decided to standardize high level communications infrastructure around the Python Server Environment (`server.py`), while migrating low level motor coordination to a specialized serial package inspired by the C++ Driver Framework (`serial_motor_controller.cpp`).

Phases of Integration

Phase 1: Deprecation of Outbound C++ HTTP Hooks

Because our control interface will be embedded directly within the preexisting React Native mobile/web app, the rover does not need to send outbound HTTP requests to itself via `libcurl`. The `motor_controller.cpp` outbound loop will be deprecated. Instead, the

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React Native front end will post commands directly to the server.py REST API entry point.

Phase 2: Preserving the 2023 ROS2 Workspace Core

We will preserve the native server.py background daemon architecture. This ensures that real world coordinate tracking and telemetry streams can still be pushed back to the client UI at high frequency using the integrated SocketIO WebSocket protocols (subscribe_odom, subscribe_map) without starving processing threads on the onboard microprocessor.

Phase 3: Hardware Bus Separation and Port ReMapping

The legacy C++ and Python serial architecture assumes the presence of two low level controllers: a wheel controller and an excavation drum controller. Our multi joint robotic arm requires an entirely independent control space. Linux rules will be appended to the microprocessor to register a new symbolic port: /dev/arduino_arm.

Scaling the UI and Web Styling Format

To ensure a clean user experience, our software modifications adhere to the styling and UI components already established in the inherited React Native application.

- **Component Reusability:** The legacy UI implements joystick interfaces and directional button grids wrapped in styling classes that handle responsive touch scaling and network timeout safety. We will clone these styling blocks to create a dedicated 'Arm Manipulation' control dashboard within the application workspace.
- **Interface Design:** The new UI module could replace the buttons with a clean, slider based joint coordinate array and a toggle switch for the end effector claw. Moving a slider will dynamically recalculate target angles in real time, grouping them into unified JSON strings before dispatching them over the network.

Forward Compatibility: Preparing for Computer Vision & Stretch Goals:

Aside from the project's primary goal. If we got the chance, we also have some stretch goal in mind: Autonomous Object Pickup. Prior teams attempted computer vision using external hardware like the NVIDIA Jetson Nano but struggled heavily with physical integration and documentation.

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To bridge this gap for future iterations, we could intentionally implement `server.py` that leaves native WebSocket channels open for streaming live camera depth matrices (`subscribe_points` and `subscribe_depth_image`). By broadcasting the downsampled XYZ camera coordinates as base64 encoded grayscale arrays directly through the web port, future teams can inject image tracking models (such as ArUco marker detection scripts) straight into this server loop. When a paver is detected, the vision script can calculate the target coordinate offsets and execute automated movements by passing coordinate payloads back through our newly refactored JSON entry points, bypassing the need for manual driver intervention.

User Interface Design Requirements

Our design for the updated User Interface (UI) is going to be based heavily on the existing UI from previous teams that worked on this project. However, the previous UI does not have the required features to fully and effectively control the robotic arm that we are planning to integrate with the rover. The following features are necessary for the updated UI to contain:

- Controls for base mounted/shoulder joint:
 - Accurate labels, visuals, and graphics
 - Button to rotate the arm forwards at the shoulder joint
 - Button to rotate the arm backwards at the shoulder joint
- Controls for middle/elbow joint:
 - Accurate labels, visuals, and graphics
 - Button to rotate the arm forwards at the elbow joint
 - Button to rotate the arm backwards at the elbow joint
- Controls for end effector:
 - Accurate labels, visuals, and graphics
 - Button to grab an object with the end effector
 - Button to release an object with the end effector

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- A small area should be reserved for any additional controls that will need to be added in future modifications, especially if future modular end effector parts require one or more addition degrees of freedom
- The buttons must function as the user would expect.
- The arm controls should match stylistically with the existing rover controls.

The following features may be useful to include:

- A slider, set of buttons, and/or text input box to adjust the sensitivity or the speed of the shoulder joint
- A slider, set of buttons, and/or text input box to adjust the sensitivity or the speed of the elbow joint
- A slider, set of buttons, and/or text input box to adjust the sensitivity or the speed of the end effector
- Buttons to perform specific combined actions, such as returning the arm to a default pose or automatically grabbing an object at a specific location

Integration of Previous UI

Luckily, the User Interface developed by previous teams is mostly usable, Instead of needing to design a new interface from the ground up, we intend to extend the existing software to incorporate the features needed for the arm, while maintaining the usability of the current features.

This interface uses HTTP requests containing JSON formatted objects to communicate between the rover's microprocessor and the laptop or other computer running the control interface. Existing controls include directional movement with tank steering, speed adjustments, and a stop button. The control interface also includes features for status monitoring, as well as access to the camera video feed.

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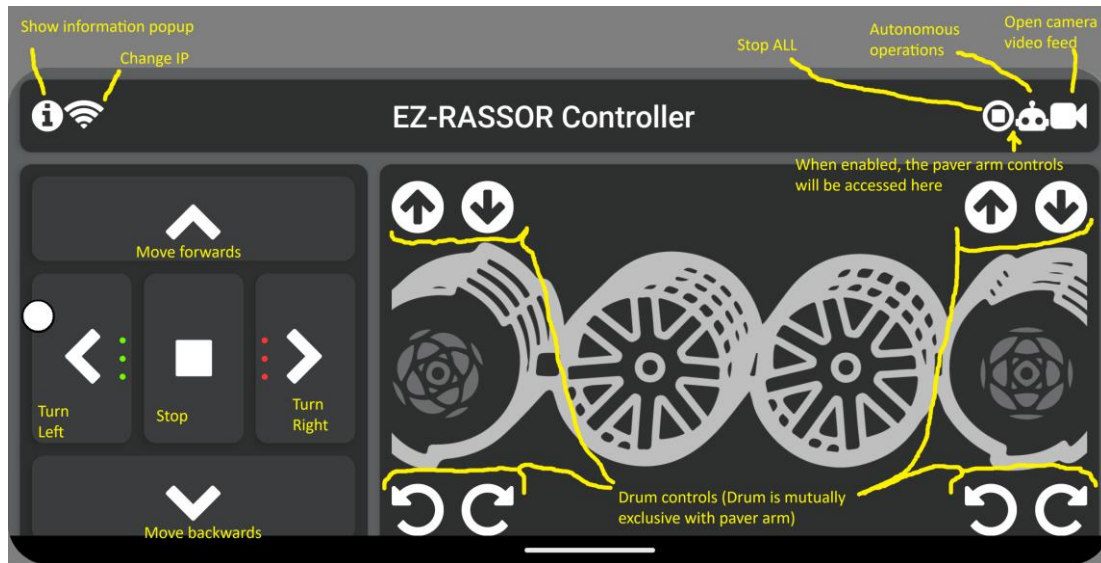


Figure 10: The Current Driving Control Panel for the Rover

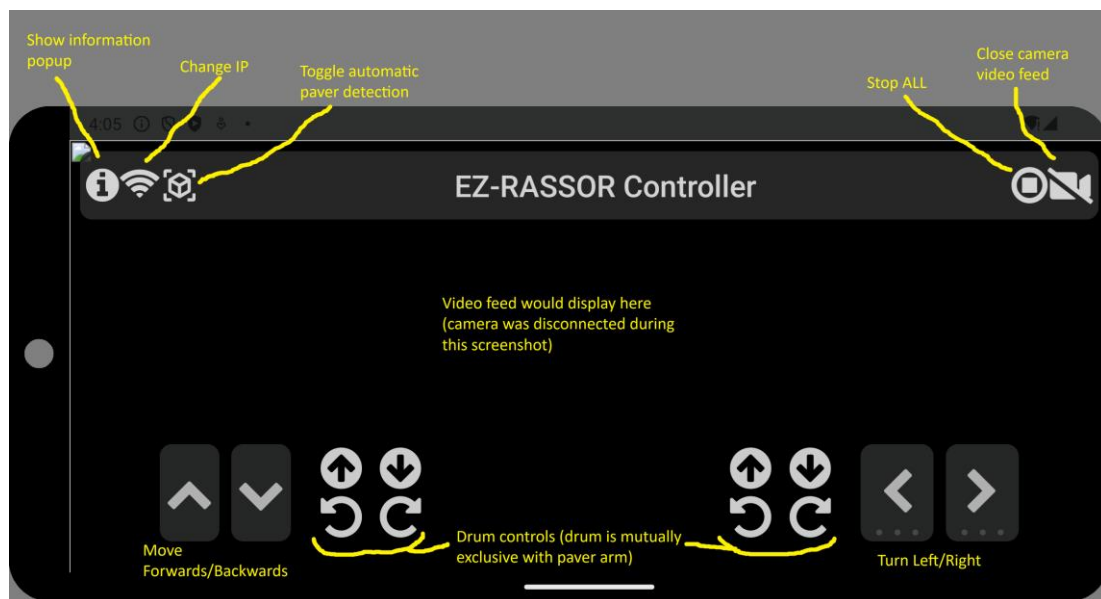


Figure 11: Drive Controller with Camera View Port

As the existing interface and its features have already been verified to function in testing by previous teams, introducing significant changes beyond the required additions has been deemed an unnecessary effort with minimal benefits.

The existing control interface does include a component for the control of a robotic arm system. The code to access this component has simply been commented out from the overall interface. However,

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this component has multiple issues and will need to be adjusted to fix those issues. Most of these issues stem from the component being developed for a different arm design than the one we are planning to use.

Most notably, the component is developed to work with an arm design with a different number of controllable joints than our design. This is especially important, as the number of controllable joints directly relates to how much information must be sent with each command, and how much information must be received as feedback.

To be more specific, our arm design is built with 3 joints:

1. Shoulder
2. Elbow
3. End effector actuator

Meanwhile the arm design the existing component is developed for has 4 joints:

1. Plate/base
2. Shoulder
3. Elbow
4. Wrist

As another issue, the graphical elements for the existing component do not represent our design for the robotic arm, and will likely be confusing if not replaced with assets that accurately reflect the current arm design. The graphical elements used in the component also do not match the rest of the UI stylistically, being untextured 3D models.



Figure 12: Example of Untextured Model

Additionally, the existing component will require some further polishing, as it currently lacks any additional features beyond the bare minimum buttons. Though, this will be relegated to a stretch goal, as these other features are not strictly necessary for functionality.

Another major consideration relating to the integration with the existing UI is trying to match the new UI elements stylistically with the existing ones. The existing interface utilizes a clean black, gray,

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and white color palette with clearly separated control panels and straightforward navigation. Most of the icons representing the rover or its parts use simple 2D symbols instead of 3D models. Using the same visual design language will make it easier for users familiar with the existing interface to use the extended version we develop.



Figure 13: Example of the 2D Symbols We Need to Replicate for the Arm

Required Materials / Tech Stack

A variety of physical materials, software, and other technological products are or will be used in the rover, the associated peripherals for the rover, and/or the development process:

- 3D printed components:
 - Two different types of plastic:
 - Polylactic Acid (PLA) - for testing prototypes and printing anything that isn't a final design.
 - Polyethylene Terephthalate Glycol (PETG) - for printing the final design and anything meant to be on the robot permanently.
 - Pavers for the arm to pick up
 - Cycloidal gearboxes for the motors
 - Arm structural casing
 - Some end effector components:
 - Contact points
 - Structural casing

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- Rover body structural casing
 - Wheels
 - Rover body shoulder plates
- Rover electronic components:
 - Brushless DC (BLDC) motors - attached to cycloidal gearboxes to move wheels, arm joints, and the end effector joint
 - Raspberry Pi and/or Le Potato microprocessor boards
 - Arduino microcontroller board(s)
 - Wires connecting the motors, microprocessors, microcontroller(s), and router to each other
 - Wires transporting power to the components
 - Battery pack
 - Power switch
 - Cooling fan
 - Cameras
- FSI-provide Router - allows the rover to connect to other devices through an isolated wireless network
- FSI-provided laptop (or another computer)
- FSI-provided regolith simulant for testing
- FSI-provided Bambu Lab X1 Carbon 3D Printer with an Automatic Material System
- Software:
 - Git - GitHub
 - EZ-RASSOR rover software:
 - Arduino scripts
 - Microprocessor programs - including network connection

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- EZ-RASSOR control software/app:
 - React Native
 - Expo
 - Android device emulator
 - Connection service
- IDEs:
 - Arduino IDE
 - Android Studio - mostly for Android device emulator
 - Visual Studio Code
- Gazebo simulator - for testing without physical rover
- AutoCAD - Computer Aided Design program
- Bambu Studio – 3D printing slicer
- Communications and file sharing:
 - Discord
 - Email software
 - Microsoft Office services
 - Google Drive

Other Considerations and Concerns

Mechanical Component Stress and Structural Linkage Deformation

The main problem we found in our study of previous versions, i.e. the 2022 arm team, is the accumulated stress on physical linkages when upscaling 3D printed mechanical structures. The original open source EEZYbotARM geometry works well at the educational desktop scale but scaling it up to meet the RE-RASSOR rover's workspace and reach constraints exponentially multiplies the structural bending moments at the primary pivots.

- **Shoulder Pivot Torque Load:** The shoulder joint is subjected to heavy static and dynamic stresses when the arm is extended to manipulate physical pavers. The design is immediately

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at risk of actuator slipping or physical joint failure when the continuous torque requirements surpass the holding capability of a single Brushless DC motor (BLDCM). To avoid this problem, while not pushing weight to unsafe levels, our team will need to integrate high ratio cycloidal or worm gearboxes.

- **Linkage Infill Optimization vs. Rigidity:** Our team will need to carefully optimize the slicing parameters in order to overcome the heavy weight limitations that caused the failure of previous iterations. You can reduce the mass by using lower internal infill percentages, such as a 15% baseline grid, but this risks introducing physical elasticity or geometric twisting in the arm links under full loading conditions. The shell along the center of the links needs to be lightweight, while the solid, dense infill rings need to be explicitly structured around the bearing mounts and motor fastening points.

Center of Gravity Shift and Rover Tipping Risk

Adding an articulating robotic arm to a very lightweight chassis creates severe, dynamic variation of the platform's center of gravity. By adding an electromechanical arm assembly to this base, the footprint dynamics of the vehicle change drastically.

- **Front Mounting Hazards:** Placing the arm assembly at the front of the vehicle shifts its center of gravity forward. When the arm reaches its maximum horizontal forward extension to pick up a paver, the tipping moment can exceed the resisting moment provided by the rear track assembly, causing the rover to pitch forward, lift its rear tracks, and lose traction.
- **Asymmetric Workspace Bounds:** A wide operational envelope (such as a 180-degree sweep) provides excellent field utility. However, moving the arm to its lateral limits (fully left or right) again pushes the center of gravity completely outside the structural safety boundary of the tracks. This creates a high risk of side tipping. To solve this, the mechanical sub-team must integrate some kind of passive or active counterweight balances. Such as positioning the heavy router or primary battery reserves toward the rear chassis bulkheads.

Spatial Interference and Chassis Structural Thresholds

Integrating new mechanical hardware onto a prebuilt rover chassis exposes immediate spatial packaging and structural integrity conflicts that must be resolved.

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- **Chassis Loading and Material Deflection:** Initial physical assessments indicate the inherited rover body panels are of relatively thin gauge and not structurally stiffened. Directly bolting the arm base to the standard top lid or thin forward bulkheads would lead to material fatigue, bending or tearing under continuous use. We are thinking of developing wide, load distributing exterior mounting plates that securely bridge across the main structural chassis seams to avoid destructive modifications to the underlying drivetrain.
- **Peripheral Access Blockage:** Mounting the arm base must not block critical maintenance pathways. Careless placement risks obstructing the primary hardware toggle switches, internal cooling fan exhausts, or the main charging umbilical ports.
- **Dynamic Cable Management:** The arm uses continuous multi axis joint rotation with high resolution magnetic encoders. Routing the wiring harness safely is a major challenge. To avoid electromagnetic interference (EMI) and signal noise, the high frequency digital signal wires from the magnetic field encoders must be clearly segregated from the high current AC phase lines powering the brushless gimbal motors. Also, the links need custom built drag chains or flexible relief loops to avoid pinching or binding of wiring during extended periods of operation.

Thermal Vulnerability and Additive Manufacturing Material Selection

The environment of the testing field requires careful selection of the thermoplastics used during additive manufacturing.

- **The Rejection of Standard PLA:** While standard Polylactic Acid (PLA) filament is highly efficient for fast prototyping and checking joint kinematics, its extremely low glass transition temperature (~55 to 60 degrees C) makes it entirely unusable for operational hardware. Direct exposure to sunlight, combined with localized heat generated by nearby electronics or brushless motor windings, can cause PLA linkages to sag, warp, or permanently deform under load. We will only be using this for the initial testing of the rover and not the final product.

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Standardizing on PETG: All finalized physical arm link, brackets and custom gear housing will be printed strictly using Polyethylene Terephthalate Glycol (PETG) for adequate structural survival. PETG has a much higher thermal threshold (~75 to 80 degrees C), better impact resistance and great chemical resistance to environmental degradation, making it an ideal material for outdoor testing fields.

Safety Protocols and Public Deployment Risk

Given that RE-RASSOR is being developed as an open source educational tool for the public to be demonstrated to students in a variety of school settings, our engineering focus must be on physical safety and unambiguous usability.

- **Low level Watchdog Timers:** A wireless web network alone can result in dropped packets or loss of signal on the NetGear Nighthawk router. If an operator commands a maximum velocity arm movement and the network drops out during the transmission, the arm could continue moving indefinitely until it hits something or breaks itself. To mitigate this risk the low level Arduino firmware will run a hardware watchdog timer. If the microcontroller fails to receive a valid command heartbeat byte within a strict window, it will override all active states and issue a full hardware halt (SerialCmd::HALT), locking the brushless motors in place.
- **System Simplicity and Accessibility:** All electrical layouts, test programs, and maintenance steps will be cleanly documented with a simplified Kanban backlog structure to keep the system very palatable for younger student audiences. Furthermore, a common “Emergency Stop” (/emergency_stop) REST route will be prominently exposed across the React Native user interface to offer an immediate safety shutoff for users during public demonstrations.

Project Workflow and Management

As of the submission of this paper, the current timeline for our project is as Follows

Week of July 6 - July 12

- Finalize initial design modifications.
- Complete CAD updates for prototype manufacturing.
- Generate print-ready files.

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- Review the assembly process.
- Identify required purchased hardware and fasteners.
- Deliverable:
 - Prototype-ready CAD package.

Week of July 13 - July 19

- Begin manufacturing prototype components.
- Print structural arm components.
- Assemble printed components.
- Resolve assembly issues as they are discovered.
- Begin preliminary testing of arm movement.
- Deliverable:
 - Partially assembled the prototype.

Week of July 20 - July 26

- Complete assembly of prototype arm.
- Integrate motors and mechanical systems.
- Mount prototype to the rover.
- Begin testing arm functionality.
- Identify design failures and improvement opportunities.
- Deliverable:
 - Fully assembled prototype.

Week of July 27 - July 31

- Conduct prototype testing.
- Evaluate lifting capability.
- Evaluate stability and reliability.
- Document performance.
- Gather sponsor feedback.
- Create Fall redesign backlog.
- Deliverables:
 - Functional arm prototype demonstrated before the end of summer semester.
 - Documented redesign recommendations for Fall development.

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Based on the results from our initial prototype, we will construct a more complete Kanban structure. If the current plan is constructed successfully, needs little editing, and passes all outlined tests, the stretch goals outlined at the end of this paper will be discussed by the team and tasks will be added accordingly to the Jira. If our current design is flawed and unable to pass the necessary testing procedures, tasks will be added to the Jira board accordingly after a plan is devised to fix the errors that caused testing failure.

As detailed in our Jira, we have chosen a Kanban style for this project. While this is not typical of Computer Science for senior design projects undertaken at the University of Central Florida, it was deemed better than its alternative Scrum for a few reasons. Scrum is a more linear style of agile development. To work, it requires a full understanding of all tasks at the beginning of the project in order to be organized into sprints. These sprints are then taken up by the team in a set timeline to ensure project completion. While it is a sound development process, it was determined that for an experimental and research-based project like RE-RASSOR, it would be less than sufficient to manage our project as our plan relies upon multiple factors working correctly to be able to commence with the next tasks. This is why Kanban was chosen. Instead of sprints, our method focuses on classifying tasks by stages of completion and doing frequent analyses of schedule to ensure that foundations are set before continuing. By using this process, we are able to add in new tasks easily when a design fails or error occurs as will likely happen due to the nature of our project and the experience of development teams before us.

Development Environment

The RE-RASSOR project utilizes several standard software tools to support collaboration, documentation, version control, and project management throughout the development process. GitHub serves as the primary repository for source code, CAD files, project documentation, and other project assets. By maintaining a centralized repository, all team members have access to the latest project files while preserving a complete history of revisions and changes made throughout development.

Microsoft Word is used for the preparation of formal project documentation, including the Preliminary Design Review, progress reports, meeting minutes, and other required deliverables. Standardized formatting within Word allows all team members to contribute to shared documents while maintaining a consistent presentation suitable for sponsor and faculty review.

Discord serves as the team's primary communication platform. Dedicated channels have been established for general discussion, technical development, sponsor communications, meeting scheduling, documentation, and software development. Organizing discussions by topic reduces unnecessary

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communication overhead while allowing project information to remain accessible throughout the duration of the project.

Jira is utilized to implement the team's Kanban workflow. Tasks are organized into project stages including planning, research, implementation, testing, sponsor review, documentation, and completion. The Kanban structure was selected due to the highly iterative nature of mechanical design and prototype development, allowing the project schedule to remain flexible as testing identifies new engineering challenges.

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Project-Specific Development Tools

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Beyond standard collaboration software, the RE-RASSOR project utilizes several specialized tools to support hardware development and systems integration.

GitHub Webhooks have been configured to communicate directly with the team's Discord server. Whenever commits, pull requests, or repository changes occur; automated notifications are posted within Discord. This integration improves project visibility by allowing all team members to monitor development activity without continuously checking the repository manually.

Microsoft OneDrive is used for long-term storage of sponsor documentation, research papers, reference material, design notes, and archived project files that are not appropriate for source control. This provides centralized access to project resources while reducing unnecessary repository size.

Autodesk AutoCAD and Autodesk Fusion are utilized for mechanical design and modification of the existing rover platform. These tools allow the team to create new components, modify inherited CAD models, perform dimensional verification, and prepare files for additive manufacturing.

Bambu Studio serves as the primary slicing software for preparing components for fabrication on the Bambu Lab X1 Carbon printer. STL models generated during CAD development are converted into printable G-code while allowing adjustment of print settings such as layer height, infill density, support generation, and material profiles.

Development of rover software is performed using Visual Studio Code and Git, allowing software modifications to be version controlled alongside mechanical development. Because the rover operates through a web-based control interface utilizing HTTP requests, software development focuses on maintaining compatibility with the existing Raspberry Pi architecture while integrating control of the robotic arm.

Feasibility, Testing, & Expansions

Overall Current Feasibility of Baseline Deliverables

Based on the team's research, sponsor guidance, and analysis of previous RE-RASSOR development efforts, we believe the baseline deliverables are feasible within the scope of this project. Existing rover hardware, prior documentation, and lessons learned from earlier arm prototypes provide a strong technical foundation that reduces development risk compared to starting from a blank design. While several previous robotic arm implementations have not progressed to long-term operational

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success, those efforts have produced valuable research that informs our design decisions and helps identify potential challenges before fabrication and testing begin.

The team also recognizes several risks that may impact development. Although members have foundational programming experience, some are still developing proficiency with skills such as computer-aided design (CAD), 3D printing workflows, and the HTTP-based communication used by the rover's control system. These learning curves introduce additional uncertainty into the development timeline. To mitigate these risks, the project will emphasize iterative prototyping, structured testing, and comprehensive documentation of design decisions, test results, and failures. As this is fundamentally a research project, unsuccessful design iterations are expected to provide valuable engineering data. Thorough documentation will ensure that both successful solutions and identified limitations can be leveraged by future RE-RASSOR teams, contributing to the long-term advancement of the platform regardless of the final outcome of this individual project.

Minimal Viable Product Testing Procedure

Testing Considerations

As the RE-RASSOR 2026 team has been recruited to a robotics-based project, we acknowledge that much of our work falls under the following three rules of robotics proposed by Isaac Asimov. Although fictional, Asimov's Three Laws of Robotics emphasize the importance of prioritizing human safety. Our testing philosophy similarly places personnel safety above all other project objectives.

- “A robot may not injure a human being or, through inaction, allow a human being to come to harm.”
- “A robot must obey orders given it by human beings except where such orders would conflict with the First Law.”
- “A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws.”

As a result during the development of our project, we have and will continue to take meticulous care to test any products we produce not only to ensure our own safety during testing and manufacturing, but also for others who may later operate and edit the deliverables we produce as outlined above in our design approach and objectives. To achieve this, we as a team have developed and multi-

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avenue approach to safety including detailed documentation of tests, sectional component tests, complete rover tests, and fixed safety procedures while working on and testing the rover and its arm.

If any of the following conditions occur, the RE-RASSOR rover and robotic arm shall be shut down immediately. Testing will be suspended until the source of the issue has been identified, corrective actions have been implemented, and the affected testing phase has been successfully repeated. These criteria are intended to ensure the safety of personnel, protect project hardware, and maintain the integrity of testing results.

- Testing shall be paused immediately if any of the following conditions are observed:
- Visible cracking, layer separation, or structural failure in any 3D printed component.
- Audible cracking, popping, or excessive deformation of PETG, PLA, or other structural components during operation.
- Visible melting, softening, or heat-related deformation of printed components.
- Any fastener, bearing, shaft, joint, or structural component becoming loose or exhibiting excessive play during testing.
- Unexpected arm movement or motion outside of the commanded path.
- Excessive vibration, oscillation, or instability that could damage the rover or robotic arm.
- Impingement, pinching, stretching, or entanglement of wires, cables, or other electronic components.
- Loss of communication between the web application, Raspberry Pi, motor controllers, or robotic arm during operation.
- Failure of any emergency stop, return-to-rest procedure, or other implemented safety feature.
- Failure of the end effector to securely retain or intentionally release a payload as commanded.
- Motor, motor controller, battery, or electronic component temperatures exceeding safe operating limits or showing signs of overheating.
- Smoke, unusual odors, electrical arcing, exposed conductors, or any indication of an electrical fault.
- The rover exhibits signs of tipping, excessive instability, or loss of balance during arm operation.
- Unexpected contact between the robotic arm and the rover chassis, electronics, testing equipment, or surrounding environment.
- Software behavior that prevents the operator from safely controlling the rover or arm.
- Any condition that presents a potential safety hazard to personnel or creates a reasonable risk of permanent damage to the rover, robotic arm, testing environment, or associated equipment.

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- Following any test suspension, the team will perform a detailed post-test analysis to determine the root cause of the failure, document observations and corrective actions, and update the project's Kanban board as necessary. Testing will only resume after the issue has been resolved, and the affected testing phase has been repeated successfully.

While this is not a complete list of errors that could occur under testing, it is a comprehensive guideline for when a given test needs to be reevaluated to ensure a safe testing environment for all.

Mechanical Testing:

Mechanical testing will determine whether our arm can perform reliably before it is tested on the full rover. We will begin with individual printed parts, then move into joint assemblies, bench testing, and rover mounted testing. This order lets us isolate fit, alignment, and stability issues before full system testing adds more variables.

The EEZYbotARM model is being used as a conceptual reference for the general arm layout. It gives us a starting point for base rotation, linked arm sections, and gripper movement. The final RE-RASSOR arm still must be redesigned around the rover chassis, the required motor layout, the mounting space, the rover height, and the shape of the 3D printed pavers. The base is expected to use a stepper motor. The remaining joints are expected to use BLDC motors. That motor layout affects testing, since the base rotates the full arm structure while the upper joints handle lifting and placement.

We will first inspect the printed components for fitness and alignment. Each part should be compared against the CAD model before motors are attached. The arm depends heavily on printed geometry, so small errors can affect the assembly. A tight rotating joint can bind during movement. A loose mounting hole can create play once the arm is under load. A warped surface can throw off alignment across multiple parts. These problems should be corrected before powered testing begins.

Early prototypes should be printed in PLA when we are checking the form and assembly. PLA is easier to print and more practical for design changes. Once the geometry is more stable, PETG should be used for parts closer to the final design. This gives us a faster early testing process while still allowing load carrying parts to be tested with a more durable material.

Motor mounting should be checked before powered motion. The stepper motor at the base and the BLDC motors at the upper joints need to align correctly with the parts they move. Poor alignment can create friction and uneven loading. Before applying power, we should move each joint by hand through its expected range.

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The base joint should be tested as its own assembly. It supports the rest of the arm and rotates the full structure, so looseness at the base will affect every movement above it. We should test for smooth rotation and consistent alignment after repeated movement. The mount also needs to spread load into the rover chassis without relying on major changes to the inherited platform. If the base shifts during testing, the upper arm results will not be reliable.

The shoulder joint should receive the most careful mechanical testing. When the arm lifts a paver, the shoulder joint carries the load from everything past that joint, including the upper arm assembly and the object being lifted. The load becomes more demanding as the arm extends away from the rover's body. Testing should begin with unloaded motion then continue with the arm structure attached. A later test should use a paver or paver substitute held at different positions. During these tests we should watch for slipping, bending or loss of position.

The elbow joint controls much of the arm's reach. It may carry less total load than the shoulder, but it still affects gripper accuracy. If the elbow has too much play, the gripper may drift away from the intended position. If the elbow binds, the motor may struggle to move the arm smoothly. Repeated elbow movement should show whether the joint stays consistent or begins to loosen after use.

The wrist and gripper area should be tested separately before full arm testing. This is the part of the arm that directly interacts with the paver. A gripper can look correct in CAD and still fail if the fingers do not close evenly or if the contact surface cannot hold the paver securely. The gripper should open far enough, close around the paver, and release without catching. It should also be tested with the paver at slightly different angles to account for imperfect rover positioning during demonstrations.

Repeatability testing is also needed. The arm should lift a paver, hold it, move it, and release it through several cycles while we observe the joints and printed parts. Repeated operation can reveal loosened screws, gear wear, joint play, and motor heat buildup. These cycles will show whether the arm can perform the expected task without constant manual adjustment.

Rover stability testing should begin before the powered arm is fully integrated. We should mount the arm structure to the rover and manually place it in several positions. The test should start with the arm empty then repeat with a paver held at the end. If the rover becomes unstable when the arm is extended, we will need to change the mounting point, weight distribution, or counterweight plan. Previous RE-RASSOR arm attempts showed that stability can become a major failure point, so this issue should be tested before the final integration stage.

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Before full rover testing, the complete arm should be tested on a fixed stand or bench setup. The arm should lift a paver, hold it in place, move it through the intended path, and release it without manual correction. Bench testing gives us a safer environment for identifying weak joints, poor alignment, unstable gripper behavior, and motor loading problems.

We will document each major test result alongside the part version, material, observed issue, and design change. Photos of failed prints, notes about joint movement, and explanations of rejected designs will make the project easier for future teams to continue.

Software Testing:

Software testing should begin before the arm is attached to the rover. The arm is expected to use the existing rover software structure, where commands are sent through the current network-based control system and integrated into the existing web application. Our first goal is to understand how existing rover commands move from the user interface to the rover controller. Once that path is clear, we can add arm commands in a way that matches the current system.

The first software task is to test the existing rover HTTP requests. This will help us understand how commands are formatted, where they are processed, and how they produce physical movement. We will trace an existing drive command from the web interface to the hardware response, to model the arm commands after the same process.

Early arm command testing should happen without a physical arm connected. At this stage, we can use logging or simple test outputs to confirm that commands are being received correctly. A command intended for the shoulder should be recognized as a shoulder command. A command intended for the gripper should not trigger rover movement. This prevents more serious issues once motors are powered.

The software should also be tested against invalid input. During development, the back end might receive repeated commands or commands sent in an unexpected order. The software should reject or ignore bad input in a predictable way. It should also avoid uncontrolled movement if a command is repeated too quickly. Since the arm is a physical mechanism, software behavior must be considered when thinking of safety.

A stop command should be tested early. The operator needs a reliable way to stop the arm during testing and demonstrations. The stop behavior should be tested while the arm is unloaded, once the motors are connected, and later while the arm is holding a paver. We should also consider what happens if

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communication is interrupted. The arm should not continue moving indefinitely after the interface disconnects.

After HTTP request testing, the software should be tested with motors disconnected from the arm structure. This allows us to confirm that each command controls the correct motor and that each motor turns in the expected direction. If the shoulder command moves the wrong motor after the arm is assembled, it could damage the structure. Motor tests before load give us a safer way to confirm wiring, motor assignment, and control logic.

The next stage should test the motors while connected to the arm on a stand. We should move one joint at a time and observe how the software behaves under real mechanical resistance. A motor might respond correctly with no load then struggle once it must move a printed arm segment. These tests will show whether movement speed, command duration, or control behavior needs changes.

Software testing should also confirm that the original rover functions still work after arm controls are added. The rover must still drive, turn, and communicate after the arm software is introduced. After major software changes, we should retest basic rover movement before returning to arm testing. This prevents us from breaking the already integrated system.

The software should be documented as it is tested. Each arm command should have a clear purpose and a clear relationship to physical action. Future teams should be able to understand which command controls the base, shoulder, elbow, wrist, or gripper without reading through the entire codebase. Known limitations should be documented while they are still fresh.

User Interface Testing:

User interface testing should focus on whether the arm can be operated clearly through the existing rover control environment. Since RE-RASSOR is meant for educational use, the controls should be understandable to people who did not design the system. The interface should let an operator control the arm without needing to understand the code for each command.

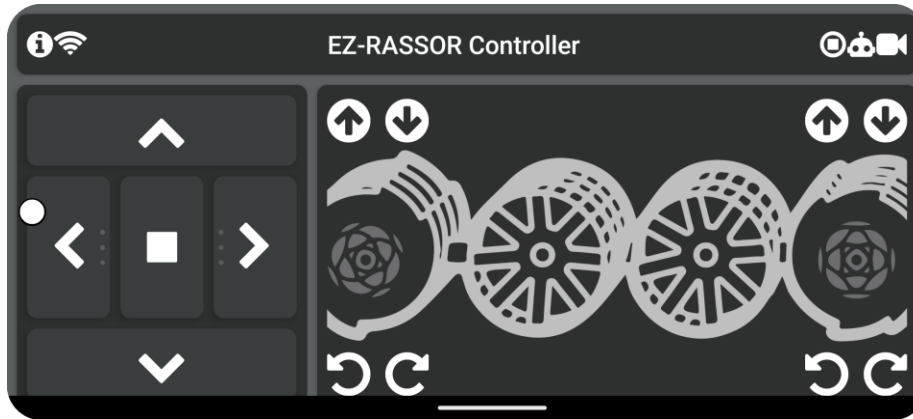


Figure 14: Example of Previous Control Design

The first UI tests should confirm that each visible control sends the correct command. This should be done before the arm is powered. If a button is labeled for the gripper, it should send a gripper command. If a control is assigned to the elbow, it should produce elbow movement once the hardware is connected. Logs can confirm the command behavior before the physical arm moves.

The interface should also be tested for clarity. Arm controls should be labeled in a way that matches the motion of the arm. Operators should not have to guess which control moves the shoulder, or which one closes the gripper. If keyboard controls are used, they should be displayed in the interface or documented near the test setup. During a demonstration, the operator should be able to tell whether they are controlling the rover or the arm.

The layout should be checked during normal operation, not only during isolated command tests. Controls that look clear on the screen may still feel awkward once the operator must move between the base, arm joints, and gripper. The most important controls should be easy to find during paver pickup. Any stop or neutral control should also be placed where the operator can reach it quickly.

Timing should be tested as part of the interface. If the operator presses and releases a control, the arm should stop when expected. If a command moves the arm by a fixed step, the step should be consistent enough for the operator to predict. Poor timing can make the arm feel unsafe or difficult to use even when the mechanical design works.

The UI should also show communication feedback. If the rover is disconnected or the arm controller is unavailable, the operator should receive some indication that the command did not complete. A simple status message would make testing easier and reduce confusion during demonstrations. Feedback is especially useful during integration testing where a failed command could come from the interface, the network connection, or the hardware.

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After the physical arm is connected, the interface should be tested during actual paver handling. The operator should use the UI to approach a paver, grip it, lift it, move it, release it, and return the arm to a resting position. This sequence will show whether the control layout works during real operation. A layout that seems reasonable during isolated joint testing might become awkward when many movements have to be coordinated.

UI testing should also include at least one person who did not build the interface. If another team member can understand the controls with minimal explanation, the layout is likely clear enough for demonstrations and future handoff. If they hesitate or press the wrong controls, the labels, layout, or documentation should be revised before final testing.

Documentation:

Documentation is especially important for RE-RASSOR due to the way the project is passed between student teams. Our arm design cannot end as a prototype that only our group understands. Future teams should be able to open the project files, understand the current design, and continue development without reverse engineering from scratch. The documentation should make the final design understandable as a system.

The mechanical documentation should clearly describe the final arm assembly. This should include the CAD files, part versions, print material, print settings, and assembly steps. If a part has a specific orientation, support requirement, or tolerance issue, that should be written down near the part it applies to. Future teams should be able to tell which files current and which ones are only used for testing. File names should also be organized clearly, so the final design is not mixed with outdated prototypes.

Any part that changed during testing should be documented with a short explanation of what changed and why the previous version was rejected. If a joint, mount, or gripper part fails during testing, that failure should be recorded and explained instead of being removed from the project history. Failed designs can still prevent future teams from repeating the same mistake. For example, if a joint was too loose, a mount flexed too much, or a gripper could not hold the paver consistently; those results should be included with the design notes.

Assembly documentation should be detailed enough for someone outside our group to rebuild the arm. The instructions should show how the printed parts connect, where each motor is mounted, how the gripper is installed, and how the arm attaches to the rover. Photos or diagrams should be included when written instructions are not enough. If a step requires careful alignment or a specific order of assembly,

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that should be stated clearly. This would reduce confusion for future teams and make repairs easier if the arm needs to be taken apart.

The software documentation should explain how the arm controls connect to the existing rover system. Each command should have a clear purpose and should match the physical movement it creates. Future teams should be able to identify which command controls the base, shoulder, elbow, wrist, or gripper without searching through the full codebase. Any setup steps, dependencies, or configuration details should also be written clearly, so the software can be run again after the project is handed off.

The control documentation should also connect the user interface to the physical arm. If a button, keyboard input, or HTTP request moves a joint, that relationship should be recorded. This will make debugging easier when a command does not produce the expected movement. It also helps future teams modify the interface without accidentally changing the wrong control behavior. If any commands are limited for safety or only meant for testing, those limits should be stated.

Testing documentation should connect design decisions to actual results. Each major test should record the setup, the part version used, the observed behavior, and the design change that followed. This applies to mechanical testing, software testing, user interface testing, and full rover integration testing. A short record of what worked and what failed will be more useful to future students than only showing the final successful demonstration.

The final handoff should also include known limitations. If the arm works only within a certain range, that range should be stated. If the gripper struggles with certain paver positions, that should be recorded. If the mount, wiring path, software controls, or counterweight concept still needs improvement, those issues should be left as clear notes for future work. Limitations should be written directly so future teams can plan around them instead of discovering them during testing.

The final folder should separate CAD files, software files, testing notes, assembly instructions, and reference material. A main README should explain where each part of the project is located and what files are needed to rebuild or run the system. This makes the project easier to build on and gives future teams a stronger starting point.

General Complete Rover Testing:

The following seven step plan was devised to test all failure possibilities of the rover under the addition of our arm design. This seven-step system will be undertaken after initial individual mechanical and software testing to reduce chances of severe malfunction during the rover's operation.

Phase 0:

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Prior to any powered testing, the mechanical frontend team will conduct a thorough inspection of the assembled arm and supporting hardware. This inspection will include verifying that all screws, bolts, nuts, spacers, and structural components are properly secured according to the assembly documentation. Wiring will be inspected to ensure connectors are fully seated, cables are properly routed, and no wiring interferes with the arm's range of motion. Electronic components, including the BLDC motors, motor controllers, Raspberry Pi, and any auxiliary hardware, will also be inspected for proper installation.

The objective of this phase is to identify assembly defects before motion testing begins. Any issues identified during inspection will be corrected prior to proceeding to Phase 1.

Phase 1:

To isolate the arm from the rover and reduce the number of variables during initial testing, the arm and its mounting platform will first be assembled independently of the rover chassis. This allows the team to verify that the arm functions correctly before integration with the complete system.

Testing will begin by verifying the operation of the end effector through multiple open and close cycles. The operator will then test the arm's vertical motion by moving the arm from its fully lowered position to its maximum extension before returning it to its lowest position. This process will verify smooth motion, identify mechanical interference, and evaluate overall stability.

Following vertical motion testing, rotational movement will be evaluated by rotating the arm through its full range of motion at multiple extension heights, including both minimum and maximum reach. The operator will observe excessive vibration, structural flex, binding, or unexpected motor behavior.

The phase will conclude by returning the arm to its predefined resting position.

Phase 2:

Following successful motion testing, the team will evaluate all implemented safety features.

Testing will include verifying that the arm safely returns to its resting position following a simulated communication failure, confirming that the end effector maintains its commanded state during unexpected interruptions to prevent accidental release of payloads, and validating any software or hardware emergency stop procedures developed throughout the project.

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Additional safety tests may be added as development progresses, and new safety features are incorporated into the system. Any safety feature that fails validation must be corrected before proceeding to payload testing.

Phase 3:

After successful completion of unloaded testing, the team will evaluate arm performance while manipulating project payloads.

Beginning from its resting position, the operator will command the arm to grasp a standard project paver using the end effector. The operator will then repeat the motion sequence established during Phase 1 while carrying the payload. This testing will evaluate joint stability, structural rigidity, motor performance, and end effector reliability under load.

The team will monitor for excessive deflection, reduced motion accuracy, motor overheating, slippage, or unintended payload release. Any deficiencies identified during loaded testing will be documented and incorporated into future design revisions.

Phase 4:

Once independent arm testing has been completed successfully, the arm will be mounted to the existing RE-RASSOR rover platform.

Initial testing will focus on verifying that arm operation does not interfere with the rover chassis, drivetrain, or existing electronics. The operator will perform basic arm movements while the rover remains stationary, including extending the arm, rotating through its operating range, and returning it to the resting position.

The team will also evaluate the structural integrity of the mounting system and monitor for excessive chassis flex or vibration resulting from arm movement.

Phase 5:

Following successful stationary testing, the rover will be operated while carrying the mounted arm.

The operator will drive the rover for predetermined distances while periodically stopping to extend and retract the arm at multiple positions. This testing will evaluate the effects of dynamic loading on rover stability, suspension behavior, and overall maneuverability.

Additional testing will include evaluating operation on uneven terrain and within simulated regolith environments to determine whether changes in the rover's center of gravity create instability or increase the likelihood of becoming immobilized. Results from this phase will guide any required balancing systems or counterweight designs.

Phase 6:

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The final testing phase will combine rover mobility with full arm functionality.

Beginning with the arm in its resting position, the rover will navigate to a designated test paver before commanding the arm to retrieve the object. The rover will then complete a predefined driving course while carrying the payload before placing the paver at a target location.

This phase represents the complete intended operation of the system and evaluates the interaction between mechanical hardware, software, communications, and rover mobility under realistic operating conditions. Successful completion of this phase will demonstrate that the system satisfies the project's Minimum Viable Product requirements.

After Test Analysis Procedure:

Following the completion of each testing phase, the team will conduct a detailed inspection and analysis of the robotic arm, rover, and all supporting hardware before proceeding to the next stage of development. Rather than immediately implementing design changes, the team will first evaluate the results of testing to identify both successful design decisions and areas requiring improvement.

The mechanical frontend team will carefully inspect all structural components for signs of degradation or failure, including cracking, warping, layer separation, stress whitening, excessive wear, or deformation of PETG and PLA printed components. Fasteners, mounting hardware, bearings, shafts, gears, and joints will be examined for loosening, excessive play, binding, or other mechanical concerns that may have developed during testing. Motor mounts and structural interfaces will also be inspected to ensure that repeated operation has not introduced fatigue or unwanted movement.

The electrical and software systems will also be reviewed following each testing phase. Wiring will be inspected for signs of wear, connector loosening, or interference caused by arm movement. Motor temperatures, controller behavior, communication reliability, and software logs will be analyzed to identify unexpected behavior, missed commands, or potential reliability concerns.

Throughout the analysis process, the team will document all observations, failures, performance metrics, and recommended improvements. Each identified issue will be evaluated to determine its severity, root cause, and potential solution before being added to the project's Kanban board for future development. This analysis will directly guide design revisions and help determine whether the prototype is ready to advance to the next testing phase or requires additional redesign before further validation.

Final Design Testing:

After all design revisions have been completed and the team is satisfied with the performance of the final prototype, a series of validation tests will be conducted to verify that the robotic arm consistently

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meets the project's Minimum Viable Product requirements. These tests are intended to evaluate the system under normal operating conditions as well as determine its practical operating limits.

The first validation will consist of a maximum payload test. The team will gradually increase the weight lifted by the arm until the predetermined design limit is reached. Throughout testing, the operator will monitor motor performance, structural deflection, joint stability, and overall rover balance to ensure that the system remains within safe operating limits. Any signs of excessive deformation, loss of stability, or unreliable operation will be documented and compared against the project's design requirements.

Following payload validation, the team will conduct a maximum runtime test. The arm will repeatedly perform its standard operating sequence over an extended period of time to evaluate long-term reliability. The objective of this test is to identify issues such as motor overheating, fastener loosening, excessive wear of moving components, degradation of 3D printed PETG components, communication failures, or software instability that may not appear during shorter testing sessions.

The team will then perform a maximum runtime under load test by repeating the same operating sequence while continuously manipulating project pavers. This testing represents a more realistic operating condition by combining extended runtime with continuous mechanical loading. Throughout the test, temperatures, structural integrity, repeatability, and overall system performance will be monitored to identify any reliability concerns before final project acceptance.

As a final validation and pending availability of Florida Space Institute facilities, the team intends to conduct testing within a simulated regolith environment similar to the intended operating conditions of the RE-RASSOR platform. Coordination for this testing will be conducted with the project sponsor to determine facility availability and scheduling. During this validation, the rover will repeat the previously established testing procedures, including mobility testing, arm operation, payload manipulation, and stability evaluation while operating on simulated lunar terrain. This environment will provide the most realistic assessment of system performance and help verify that the robotic arm integrates successfully with the existing rover platform under representative operating conditions.

Upon completion of all validation testing, the team will perform one final post-test inspection and document all performance metrics, observed wear, design limitations, and recommendations. These results will be incorporated into the project's final documentation and provided to future RE-RASSOR teams to support continued development of the platform.

Stretch Goals & Their Individual Feasibility

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The primary objective of the project is to build a rover mounted arm that can lift and manipulate 3D printed pavers through the existing control system. Stretch goals should be considered after the baseline arm is functioning and approved by the sponsor. Each stretch goal should be judged by the amount of redesign it would require, the risk it creates for the required deliverable, and the amount of value it adds to the RE-RASSOR platform. A stretch goal is more realistic if it builds on hardware or software that already exists in the baseline design. A stretch goal is less realistic if it depends on several unfinished systems working at the same time.

Autonomous Object Pickup:

Autonomous object pickup would allow the rover to identify a paver and pick it up without direct manual control. The rover would need to detect the object, estimate its position relative to the arm, move into a usable position, and run the pickup sequence. This would make the rover more interesting for educational demonstrations by connecting sensing, control, and physical manipulation.

This stretch goal has low feasibility for the current project scope. It depends on several systems being reliable before autonomy can be attempted. The camera would need a clear view of the paver from a useful angle. The software would need to detect the paver under changing lighting conditions. The rover would need to position itself close enough for the arm to reach the target. The arm would also need repeatable movement, so the pickup sequence produces the same result each time. If any of those pieces are inconsistent, autonomous behavior will be difficult to trust.

The largest issue is that autonomous pickup depends on mechanical repeatability. A human operator can correct small alignment problems during manual control. An autonomous routine would have to rely on the arm moving in a predictable way. If there is play in the shoulder, flex in the mount, or inconsistent gripper contact, the software may correctly detect the paver and still fail during pickup. For that reason, autonomous pickup should not be attempted until the arm can already perform the paver task reliably under manual control.

This goal could still be supported through design choices made during the baseline project. We can leave room for a future camera mount, document the arm's range of motion, and organize the software so future commands can be added cleanly. If the current team has extra time, a limited version of the goal could be explored through basic image viewing or simple object detection tests. Full autonomous pickup should remain in long term work for a future RE-RASSOR team.

Additional Wrist Joint for Rotation:

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An additional wrist joint would allow the gripper to rotate along another axis. This could help the arm approach objects from more angles and may support future end effectors. A fixed gripper may work for pavers if the rover approaches them consistently. More complex objects may require the gripper to rotate before it can make contact correctly.

This stretch goal has medium feasibility. It extends the current arm concept, but it affects the most sensitive part of the design. Any added weight near the wrist increases the load carried by the shoulder and elbow. The joint would also require another motor connection, more wiring through the arm, new control behavior, and more CAD work. These changes are manageable if the baseline arm has enough weight margin. They become risky if the shoulder joint is already near its practical limit.

The main benefit of this goal is improved end effector control. A wrist rotation joint could make the gripper more useful when the rover does not approach the paver from the ideal direction. It could also make future attachments more practical, especially tools that need to face a certain direction before contacting the ground or an object. For example, a scoop or collection tool may need a different orientation than a simple paver gripper.

The decision to add a wrist joint should come from baseline testing. If the gripper can pick up and place pavers consistently without extra rotation, the wrist joint should remain a future improvement. If testing shows that the gripper struggles mainly due to an approach angle, then wrist rotation becomes more useful. This goal should only be pursued after the shoulder, and elbow joints show that they can handle the added load.

Interchangeable End Effector Attachments:

Interchangeable end effectors would allow the arm to use different tools without redesigning the full mechanism. The baseline design will focus on a gripper for 3D printed pavers. A modular connection would let future teams create alternate grippers, scoops, collection tools, or classroom demonstration attachments.

This stretch goal has high feasibility since it builds directly on the required gripper work. The arm already needs an end effector, so the connection point can be designed with future replacement in mind. The main challenge is making the connection strong enough that it does not loosen during lifting. The attachment system should not add unnecessary weight to the wrist or weaken the baseline gripper. A simple and rigid connection would be more valuable than a complicated system that is difficult to print or assemble.

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This goal also fits the long-term purpose of RE-RASSOR. The rover is meant to be reused by future student teams, so a modular end effector interface would make future work easier. A later group could design a new attachment while keeping the same arm structure. That would reduce the number of redesigns required for future projects and make the arm more useful as an educational platform.

The simplest version of this goal would be a standardized mounting pattern at the end of the wrist. The paver gripper would use that pattern, and future tools could be designed around the same interface. The attachment does not need to be complex to be useful. A reliable screw based or keyed connection could give future teams enough structure to build from without adding unnecessary mechanisms to the current prototype.

This should be one of the first stretch goals considered after the baseline arm works. It adds value without requiring object detection, autonomy, or major changes to the rover software. It also encourages better documentation, since the mounting interface would need clear dimensions and assembly instructions for future use.

Camera Integrated into the End Effector:

A camera near the end effector would help the operator see the paver and gripper during pickup. The main rover camera may not show the contact point clearly when the arm is close to the ground. A closer view could make manual operation easier and reduce failed pickup attempts during demonstrations.

This stretch goal has high feasibility if the camera is first used as a manual viewing aid. Camera placement would be the main design issue. The camera should show the gripper's contact with the paver without blocking the gripper or sitting in a location where it could hit the ground. Wiring would also need to move with the arm without catching on the joints. The camera mount should stay lightweight, so it does not add unnecessary load to the wrist.

The camera would also support testing. During early paver pickup attempts, a close view of the gripper could help us understand why a lift succeeds or fails. If the paver slips, the camera may show whether the issue came from poor alignment, weak contact, or an awkward approach angle. This would make the camera useful even before object detection is attempted.

The difficulty increases if the camera is expected to support object detection. A camera feed for a human operator is much simpler than a vision system that identifies pavers and supports autonomous pickup. Object detection would need software that can recognize the paver and produce information the rover can use. That work depends on camera placement, image quality, lighting, and repeatable arm behavior.

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A practical approach would be to mount and test the camera for operator viewing first. Once that works, the same camera location could be evaluated for future object detection. This keeps the stretch goal realistic while still moving the project toward future autonomy. If time is limited, the manual camera view should be prioritized over computer vision.

Complex Counterweight System:

A counterweight system could help address rover stability while the arm is extended. The preliminary design considers a rear storage bin that could also act as a counterweight. This concept becomes stronger if the added mass has a function, such as storing collected pavers or supporting future regolith style demonstrations.

This stretch goal has medium feasibility. A simple rear storage bin or passive counterweight could be designed without major software changes. A dumping or collection mechanism would require more mechanical design and more testing. It could also add enough weight to affect driving performance. We should decide whether a counterweight is necessary based on stability testing rather than adding one by default.

The main value of the counterweight system is stability. If the arm causes the rover to tip or lift a wheel during paver handling, a rear counterweight may help keep the rover balanced. A storage bin could make that added weight more useful by giving it a secondary purpose. Instead of adding unused mass, the rear section could support collection or demonstration tasks.

The main risk is overcomplicating the rover. Extra weight may help balance the arm but it can also make the rover harder to drive. A more complex rear mechanism could introduce new failure points and take time away from the arm itself. A passive counterweight or fixed storage bin is more realistic for the current scope. A moving dump mechanism should be treated as future work unless the baseline arm is finished early.

The counterweight concept is a good topic for future teams if the current semester does not leave enough time. A storage or regolith collection system would connect well to the larger RE-RASSOR purpose, but it should not distract from completing the arm. The best outcome for our team may be documenting the concept clearly, recording stability test results, and leaving enough design information for future development.